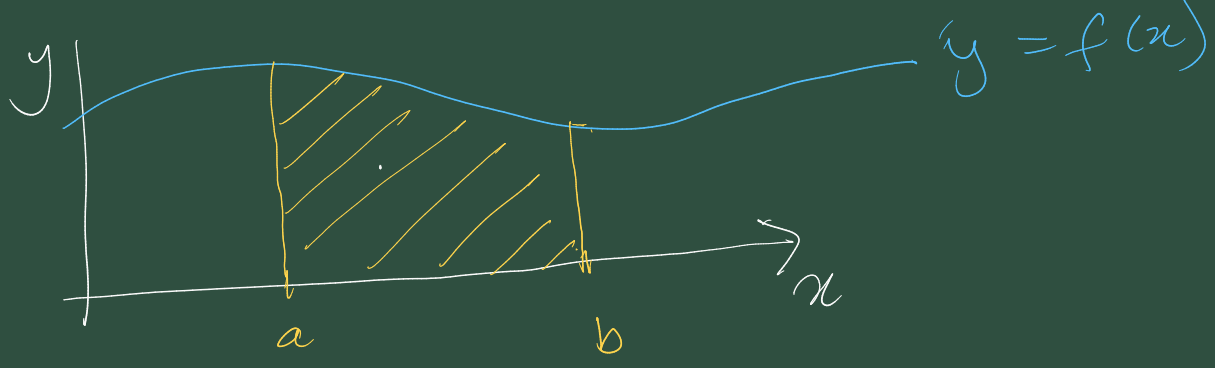
- Integration.
- another stationary points example.

Two types of multi-variable integral.
(mostly considering 2-variables)

Single variable case
- anti-differentiation.
- definite integral.

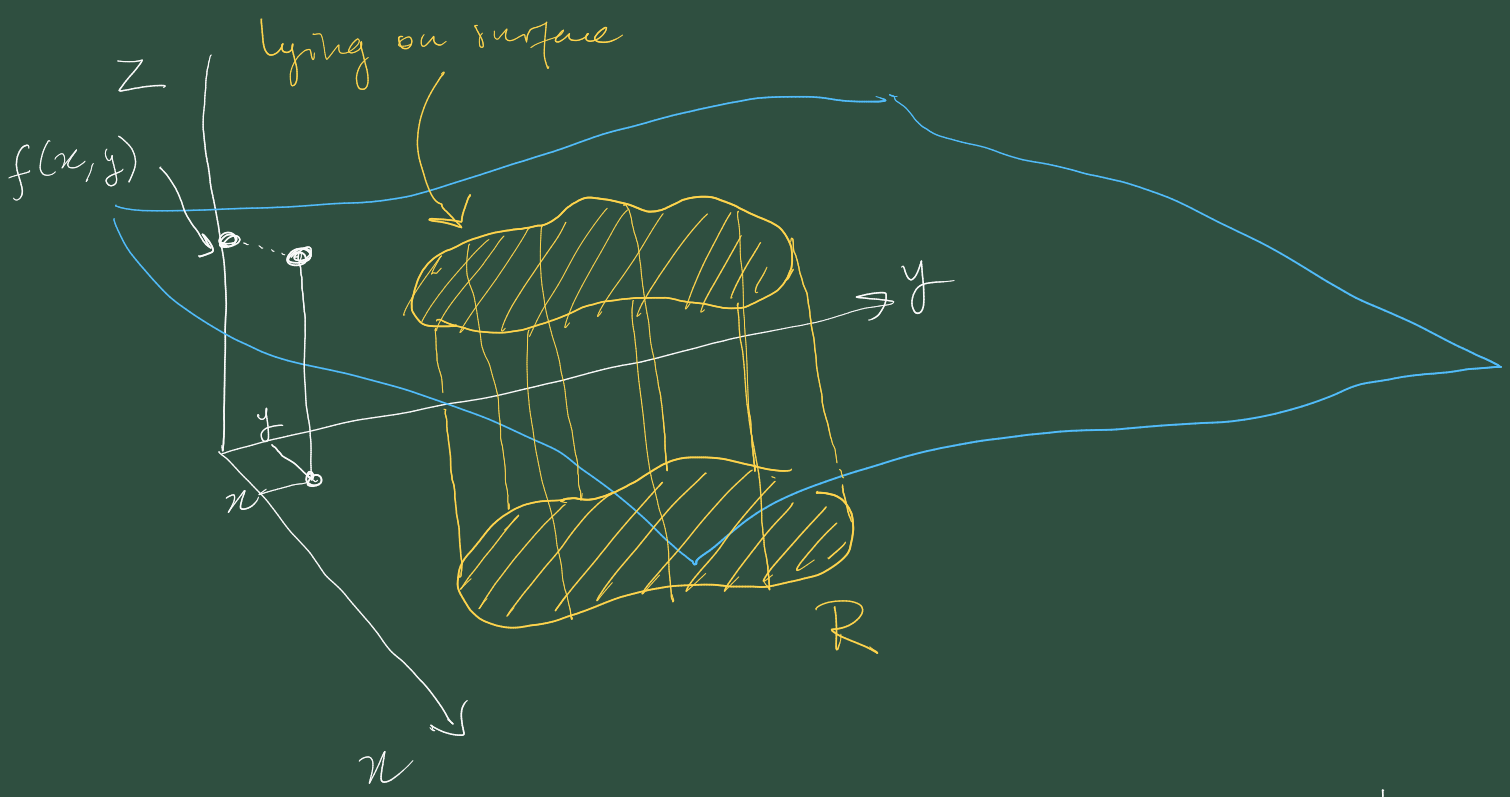


Definite integral.

$$I = \int_a^b f(x)\,dx$$

gives the area between the ~~curve~~ curve, and x-axis, between a and b.

This concept can be generalized to the setting of a surface $z = f(x, y)$ defined by a 2-variable function f.



Given a subset R of the plane we consider the volume lying over R, bounded by R and its projection to the surface $z = f(x,y)$. This will be given by the double integral (1)

$$I = \iint_R f(x,y)\, dx\, dy$$

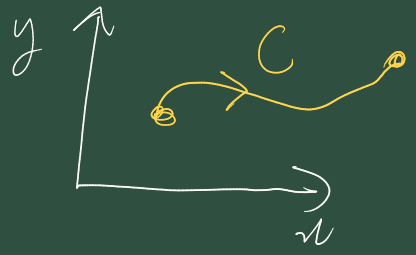
We'll discuss these next week.

---

(2) The second type is a, so called, "path integral" or "line integral" of a linear differential form L,

$$L = P(x,y)\, dx + Q(x,y)\, dy$$

with the integral taken over a path/arve C in my plane,



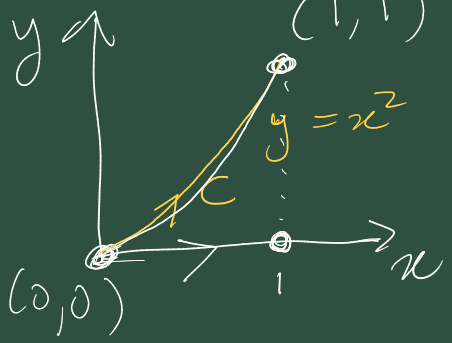
and expresed as

$$\int_C L = \int_C \left(P(x,y)\,dx + Q(x,y)\,dy \right)$$

Example: Consider the lin. diff. form

$$L = \underbrace{10x^2y}_{P(x,y)}\,dx + \underbrace{(3x+2y)}_{Q(x,y)}\,dy$$

integrate this along the curve $C: y = x^2$ from $(0,0)$ to $(1,1)$



To evaluate the integral

$$I = \int_C 10x^2y\,dx + (3x+2y)\,dy$$

we use the specification of the arve to translate I into a familiar single-variable integral

$$\boxed{y = x^2} \Rightarrow \boxed{dy = 2x\,dx}$$

$$I = \int_C 10x^2 x^2\,dx + (3x + 2x^2)\,2x\,dx.$$

$$= \int_{x=0}^{x=1} \left(10x^4 + 4x^3 + 6x^2\right) dx$$

$$= \left[2x^5 + x^4 + 2x^3\right]_0^1$$

$$= 5$$

Theorem

1. Integration is linear

So For $\alpha, \beta \in \mathbb{R}$.

$$\int_C \alpha L_1 + \beta L_2 = \alpha \int_C L_1 + \beta \int_C L_2$$

or in particular.

$$\int_C P(x,y)\,dx + Q(x,y)\,dy$$

$$= \left(\int_C P(x,y)\,dx\right) + \left(\int_C Q(x,y)\,dy\right)$$

Remember. x, y not independent here as the point (x,y) traverses the curve C.

2. Reversing the direction.

Consider C_1, C_2, where C_2 is the same curve as C_1, but traversed in the opposite direction



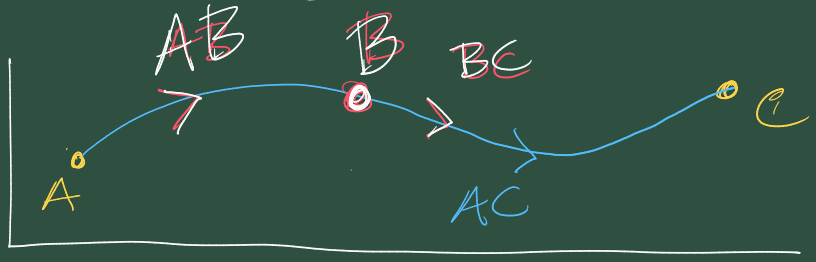
then $\int_{C_1} L = -\int_{C_2} L$

indeed we will often write $C_2 = -C_1$

Compare this with 1-variable property.

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

3. Subdividing paths.



Indeed, we can write:

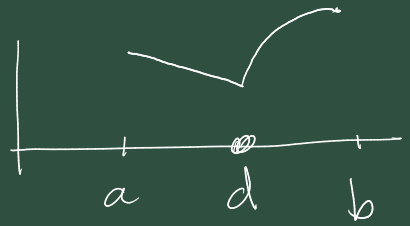
$AC = AB + BC$

concatenation of curves.

$$\int_{AC} L = \left(\int_{AB} L\right) + \left(\int_{BC} L\right)$$

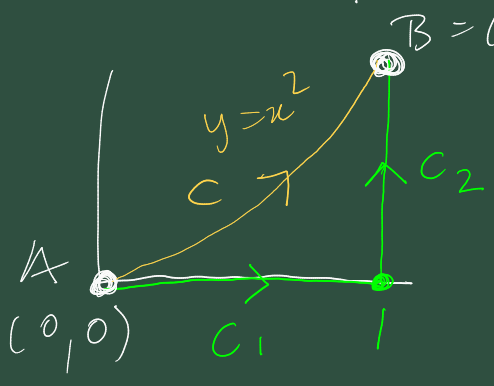
Compare with 1-variable concept

$$\int_a^b f(x)\,dx = \int_a^d f(x)\,dx + \int_d^b f(x)\,dx$$



Example 2 Same L as before.

$L = 10x^2 y\,dx + (3x+2y)\,dy$ and same start and end points $A=(0,0)$, $B=(1,1)$ but along a different path, $C_1 + C_2$



C_1 : $y=0$, $x: 0 \longmapsto 1$.

C_2 : $x=1$, $y: 0 \longmapsto 1$

So, lets evaluate.

$$\int_{C_1+C_2} L = \int_{C_1} L + \int_{C_2} L$$

on C_1 : $y=0$, $dy=0$, $x: 0 \longmapsto 1$.

on C_2 : $x=1$, $dx=0$, $y: 0 \longmapsto 1$

$$= \int_0^1 10x^2 \cdot 0 \cdot dx + (3x+0)0 + \int_0^1 (3+2y)\,dy$$

$$= 0 + \left[3y + y^2\right]_0^1$$

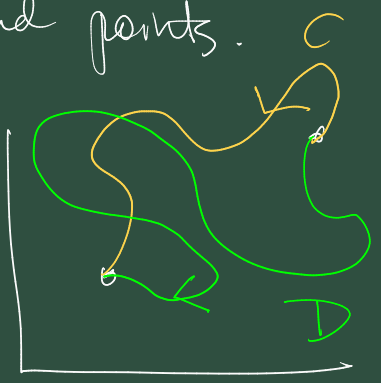
$$= 4.$$

So $\int_{C_1+C_2} L = 4$, but before we had $\int_C L = 5$

This illustrates the behaviour known a path-dependence of a linear diff. form L, the typical behaviour.

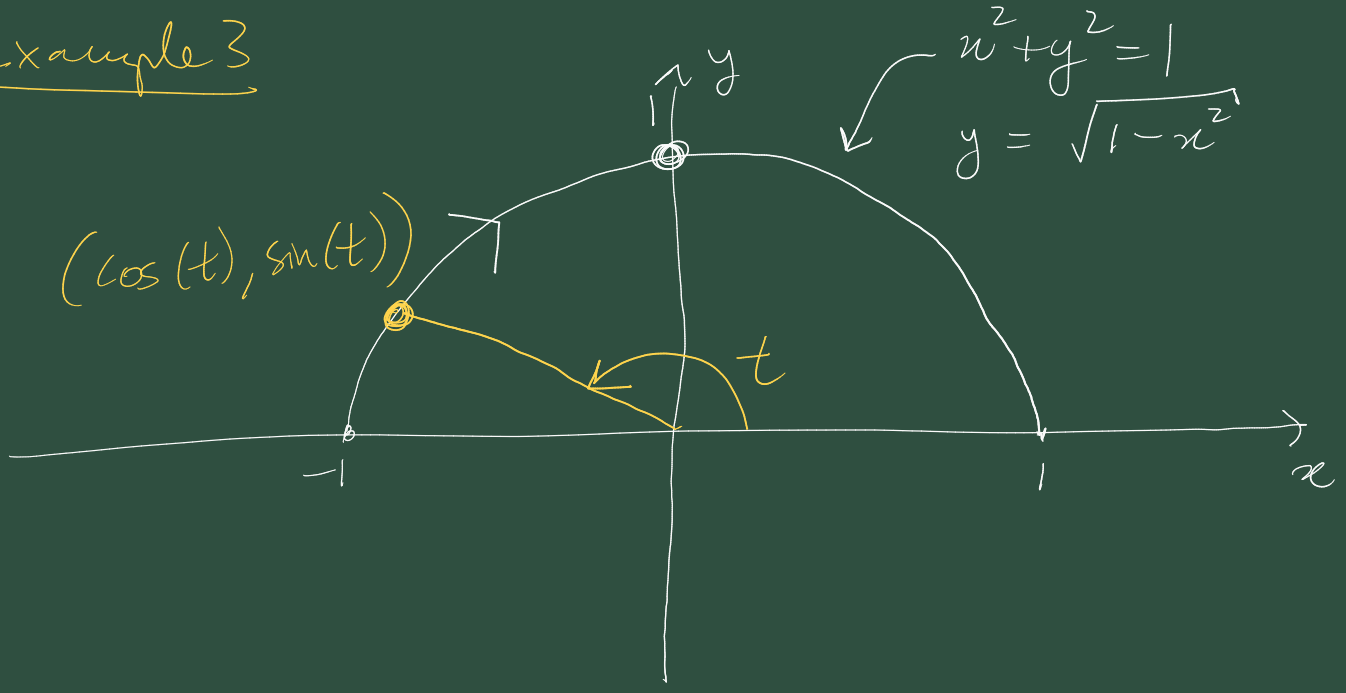
But for certain special lin. diff. forms they have a property known as path independence. where $\int_C L = \int_D L$, for any paths C, D with

~~start~~ with the same start and end points.



(Continue at 12:00)

Example 3



Find and classify stationary points of

$$g(x,y) = -2x^4 + 4x^2e^y - 4e^{4y}$$

Obtain 1st and 2nd order partial derivatives

$$\frac{\partial g}{\partial x} = -8x^3 + 8xe^y$$

$$\frac{\partial g}{\partial y} = \boxed{4x^2e^y} \quad \boxed{-16e^{4y}}$$

$$\frac{\partial^2 g}{\partial x^2} = -24x^2 + 8e^y$$

$$\frac{\partial^2 g}{\partial y^2} = 4x^2 e^y - 64e^{4y}$$

$$\frac{\partial^2 g}{\partial x \partial y} = 8xe^y.$$

where each derivative was obtained by applying the basic principle of holding other variable(s) constant when differentiating with respect to one variable.

Stationary points (a,b) are the simultaneous solutions to $\frac{\partial g}{\partial x} = 0$ & $\frac{\partial g}{\partial y} = 0$

i.e. $\frac{\partial g}{\partial x} = -8x^3 + 8xe^y = 0$ & $\frac{\partial g}{\partial y} = 4x^2 e^y - 16e^{4y} = 0$

$\frac{\partial g}{\partial x} = 0$

$\Leftrightarrow -8x^3 + 8xe^y = 0$

$\Leftrightarrow -8x(x^2 - e^y) = 0$



$\Leftrightarrow x = 0$ OR $x^2 - e^y = 0$

$\Leftrightarrow x = 0$ OR $x^2 = e^y$

First, assume $x = 0$, then $\frac{\partial g}{\partial y} = 0$

$\Leftrightarrow -16e^{4y} = 0$

which is always false

So there are no stationary points of the form $(0, y)$

Secondly, assume $x^2 = e^y$

then $\frac{\partial g}{\partial y} = 0 \iff 4e^y e^y - 16e^{4y} = 0$

$\iff 4e^{2y} - 16e^{4y} = 0$

$\iff e^{2y}(1 - 4e^{2y}) = 0$

$\iff e^{2y} = 0$ (never) OR $(1 - 4e^{2y}) = 0$

$\iff e^{2y} = \frac{1}{4}$.

$\iff e^y = \frac{1}{2}$, $e^y > 0$, so $x \neq -\frac{1}{2}$

$\iff \boxed{y = \ln\left(\frac{1}{2}\right) = -\ln(2)}$

and so there are two stationary points here.

$$\left(\frac{1}{\sqrt{2}}, -\ln(2)\right), \quad \left(-\frac{1}{\sqrt{2}}, -\ln(2)\right)$$

$x^2 = e^y = \frac{1}{2}$, $x = \pm\frac{1}{\sqrt{2}}$.

Classified using Hessian determinant

$$D = \frac{\partial^2 g}{\partial x^2}\frac{\partial^2 g}{\partial y^2} - \left(\frac{\partial^2 g}{\partial x \partial y}\right)^2$$

$= \ldots 16 > 0$ at $\left(\pm\frac{1}{\sqrt{2}}, -\ln(2)\right)$

and $\frac{\partial^2 g}{\partial x^2} = -8 < 0$, at both points ☐

So these are both local maximums.

Check / verify with Matlab.

The critical points of an example function.

```
clear
```

```
syms x y
```

```

```

```
f(x,y) =-2*x^4 + 4*x^2*exp(y) - 4*exp(4*y)
```

f(x, y) = $-2\,x^4+4\,\mathrm{e}^{y}\,x^2-4\,\mathrm{e}^{4\,y}$

```
fx = diff(f,x)
```

fx(x, y) = $8\,x\,\mathrm{e}^{y}-8\,x^3$

```
fy = diff(f,y)
```

fy(x, y) = $4\,x^2\,\mathrm{e}^{y}-16\,\mathrm{e}^{4\,y}$

```
fxy = diff(fx,y)
```

fxy(x, y) = $8\,x\,\mathrm{e}^{y}$

```
fxx = diff(fx,x)
```

fxx(x, y) = $8\,\mathrm{e}^{y}-24\,x^2$

```
fyy = diff(fy,y)
```

fyy(x, y) = $4\,x^2\,\mathrm{e}^{y}-64\,\mathrm{e}^{4\,y}$

```
sol = solve([fx(x,y)==0,fy(x,y)==0],[x,y],"Real",true)
```

```
sol =
struct with fields:
    x: [2×1 sym]
    y: [2×1 sym]
```

```
sol.x
```

```
ans =
```

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

```
sol.y
```

```
ans =
```

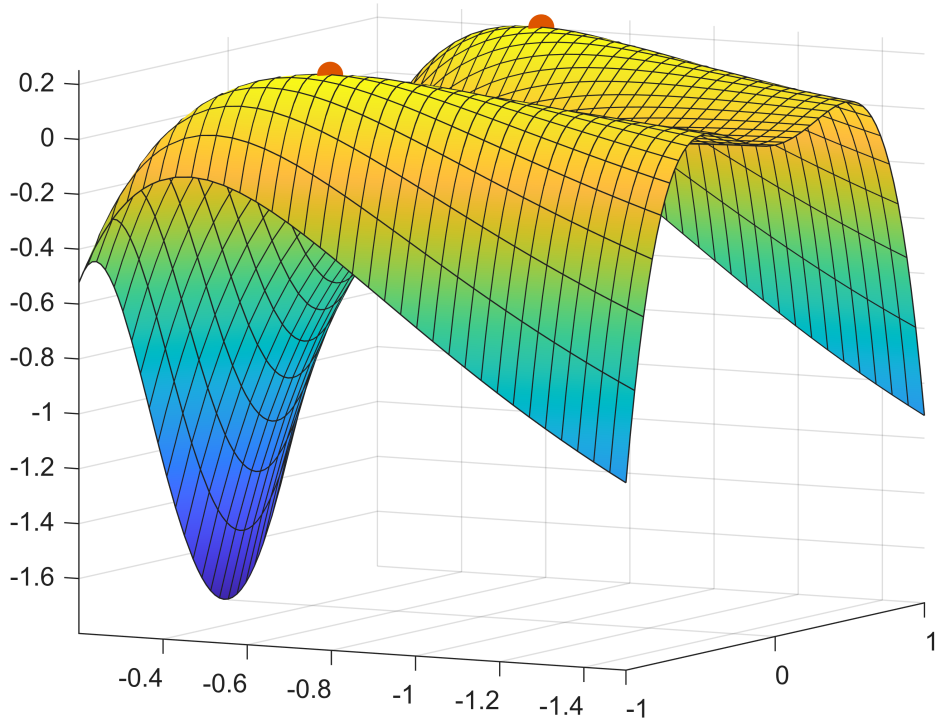
$$\begin{pmatrix} -\log(2) \\ -\log(2) \end{pmatrix}$$

```
for i=1:2
    disp([sol.x(i),sol.y(i)])
end
```

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\log(2) \end{pmatrix}$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\log(2) \end{pmatrix}$$

```
fsurf(f(x,y),[-1,1,-1.5,-0.2])
hold on
plot3(sol.x,sol.y,f(sol.x,sol.y),'.','MarkerSize',30)
hold off
```


0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-1.4
-1.6
-0.4
-0.6
-0.8
-1
-1.2
-1.4
-1
0
1